Individual Data Plan in Virtual Network Operation: A Proactive Matching Approach

Zihan Wang†, Yuan Zhuang†, Zhi Wang* and Xin Wu†
Graduate School at Shenzhen, Tsinghua University
†{wzh16, zhuangy16, wux17}@mails.tsinghua.edu.cn, *wangzhi@sz.tsinghua.edu.cn

***Abstract*—High-quality cellular wireless connection has become a necessity for today's internet users, who however, have to pay high prices when they use conventional ISPs' networks. To alleviate this pain, a new type of "mobile virtual network operators," which purchases data plans from multiple ISPs and resells them to individuals in an on-demand way, has emerged. A great challenge for such virtual network operators is to determine how to "match" available data plans to dynamically changing users' demands in an intelligent and instant manner. To this end, we propose CAPITAL, a proactive resource allocation framework based on behavior-driven individual cellular data usage prediction. First, based on measurement studies of a real-world virtual network provider over 2 years covering 3 million users, we reveal the dynamical cellular data demands of users, and the spatial, temporal and behavioral factors that affect individuals' data usage; Second, we propose a context-aware prediction model based on wide & deep network, to capture individual data usage; Third, we formulate the data plan allocation as an assignment problem, and use a shortest augmenting path algorithm to solve it efficiently. Our trace-driven experiments show that our design can reduce the cellular data cost by 78.6% for the network operator, while retaining the same user experience level.**

I. Introduction

High-quality cellular wireless connection has become a must for today's internet users. According to a mobile report[1], more than 1/4 of global users are using mobile devices only and mobile usage accounts for over half of all "digital minutes" in people's daily life. This phenomenon is also observed on a higher level: according to Cisco whitepaper, the traffic from wireless and mobile devices will account for more than 63% of total IP traffic by 2021 [1].

A significant pain point to cellular users is that they usually have to pay high prices for the data plans when they use conventional ISPs' services [2]. To alleviate this pain, a new type of "mobile virtual network operators" such as Google Project Fi, which purchases network capacity from multiple ISPs and resell these resources to individuals in an on-demand way (e.g., through software authentication), have emerged. The network capacity bought from ISPs can be in diversified forms such as *data plans*. Compared with conventional ISPs, the key solution for such network operators is to match their data plans to different users in a more intelligent and flexible manner.

Though the idea seems intuitive, designing good strategies for the virtual network operators to allocate data plans from conventional ISPs and re-distribute them to users is challenging in nature.

Fine-grained data plan allocation for dynamically changing user mobile data demand. Different from fixed monthly data plan in conventional ISPs, instant (i.e., usually daily) data plan is provided to users by certain virtual network operators. It is challenging because virtual network operators have to perform the distribution of relatively long-term data plans originally purchased from conventional ISPs, to short-term data requirements. The mismatch of the time intervals requires the virtual network operator to be able to perform *proactive* allocation over a large number of users with changing demands.

Understanding users' mobile data usage is critical for proactive data plan distribution, which is challenging because it is affected by many confounding factors. In our study, we observe that many factors affect the daily cellular data usage for a user, including when and where the user is [3], user mobility and user preference for applications. Generalized linear regression models are not adequate for leveraging such heterogeneous factors [4]. Developing a good machine/deep learning model for incorporating these factors is thus non-trivial.

Extremely large exploration space in practical virtual data allocation. The optimal data plan allocation requires a huge searching space even for small virtual network operators with thousands of users, over time. Such large exploration space have challenged conventional searching algorithms, especially when "instant" demands are to be served. More efficient solutions are in demand to enforce such individual data plan allocation.

To address the challenges above, we propose CAPITAL, a **c**ontext-**a**ware **p**roactive **i**nstan**t** **al**location framework for virtual cellular operation. As illustrated in Fig. 1, our design makes use of personal cellular data usage prediction, and allocate data plans efficiently in an on-demand manner. Our contributions can be summarized as follows.

▷ First, based on measurement studies of a real-world virtual network operator over 2 years covering 3 million users, we reveal the dynamical cellular data demands of individuals. The daily cellular data consumption can change by 89x in two consecutive days, and by 136x in different locations, for the same users, on average. We also observe that factors including time, location, mobility and online behaviors all affect the daily cellular data usage, with correlation coefficients of 0.35, 0.34, 0.33 and 0.61, respectively.

*Zhi Wang is the corresponding author.

[1]https://www.comscore.com/Insights/Presentations-and-Whitepapers/2017/The-Global-Mobile-Report

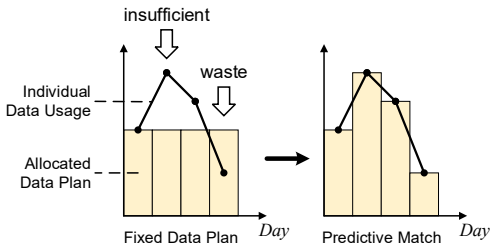


Fig. 1. Predictive fine-grained data plan allocation reduces costs of mismatch between data capacities and individual demands.

TABLE I
DATA FIELDS AND DESCRIPTION OF OUR DATASET

Data Field	Description
Timestamp	Date time when this record is pushed into database.
User ID	The encrypted and unique identifier of each user.
IP address	The destination IP address of this connection.
Domain name	The corresponding domain name to the destination IP address.
Data usage	Data traffic usage during this connection, in bytes.
Cell information	Mobile Country Code & Mobile Network Code & Local Area Code & Cell ID of the base station.

▷ Second, we propose a context-aware prediction model based on wide & deep network structure [5], to capture individual data usage. On the one hand, we apply a shallow neural network to capture the intrinsic characteristics of the daily cellular data usage; on the other hand, we also incorporate context features like user mobility into a Long-Short Term Memory (LSTM) network [6] to learn the inherent sequential pattern. Our evaluation on 500K users shows that the context-aware prediction framework outperforms conventional regression model by 37.1%.

▷ Third, we formulate the individual cellular data plan allocation as an assignment problem, and use a shortest augmenting path algorithm [7] to efficiently find the optimal data plan and user matchings. Compared to the conventional algorithm, our method can save 96% of time in practical usage. The trace-driven experiments show that our framework can reduce the data usage cost by 78.6%, while maintaining the same level of user experience (satisfaction of cellular data allocated to users).

The rest of the paper is organized as follows. We present the measurement results and analysis in Section II. We present the individual cellular usage prediction based on context-incorporated wide & deep model in Section III. We formulate the cellular data plan allocation problem and present an instant solution in Section IV. We present our trace-driven experiments to show the performance of our framework in Section V, and present related works in Section VI. Finally, we conclude the paper in Section VII.

II. MEASUREMENT OF INDIVIDUAL CELLULAR USAGE

In this section, we carry out large-scale measurement studies on how individuals use cellular data and how spatiotemporal factors and user behaviors affect it.

A. Dataset Description

Our measurement studies are based on a large-scale real-world dataset collected by our industrial partner, a mobile virtual network operator. The virtual network operator purchases data plans from various ISPs in different countries, and then provides **daily** subscription of wireless network service to customers, which is actually a re-distribution of data plan resources. Customer receives a portable Wi-Fi hotspot device to obtain network accessibility, which connects to different cellular networks provided by the data plans it subscribed[2].

The dataset in our measurement studies has a time span of 2 years, between July 2015 and June 2017; and it covers 3 million users using cellular data in 150 countries and includes 7 billion records. Each record is collected and uploaded every 5 minutes per user per connection, and contains the following information, as listed in Table I: 1) a timestamp when the record is logged; 2) an encrypted user identifier which is unique in the system; 3) the destination IP address and corresponding domain in the connection; 4) data usage during the connection; and 5) cell information, including mobile country code, mobile network code, local area code and cell ID of the base station.

In our study, we need to investigate the fine-grained cellular data usage of individuals; so we aggregate the above connection records in daily granularity, i.e., all records of the same user in the same day are aggregated. In this way, we have the daily cellular data usage for each user. Next, based on the sample data extracted from $100,000$ randomly selected users who contribute more than 3 days of connection records, we present how time, location, user mobility and use of mobile applications are correlated to individuals' cellular data usage.

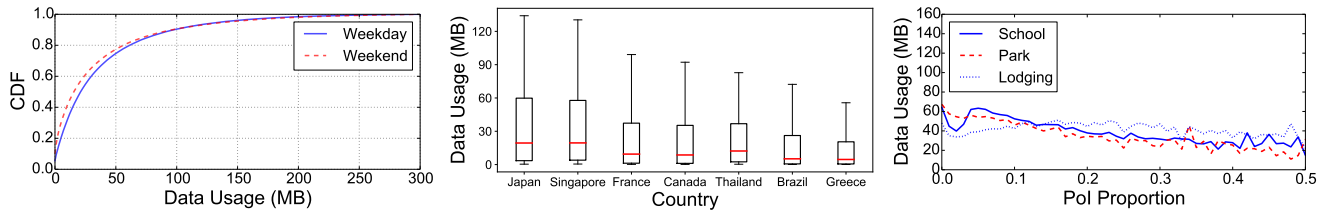
B. Time versus Cellular Data Usage

First, we study the correlation between time and the cellular data usage, and meanwhile show the distribution of individual data usage. We use the "weekday" and "weekend" to represent the time, and the data usage in megabytes as the cellular data usage. As illustrated in Fig. 2(a), we plot the distribution of cellular data usage of the selected users. We observe that 90% of users consume less than 100 megabytes of data every day while the remaining 10% consumes a relatively large amount of data causing a long-tailed distribution. Furthermore, people consumes slightly less data on weekends than on weekdays and this result suggests that there can be certain correlation between time and cellular usage.

C. Location versus Cellular Data Usage

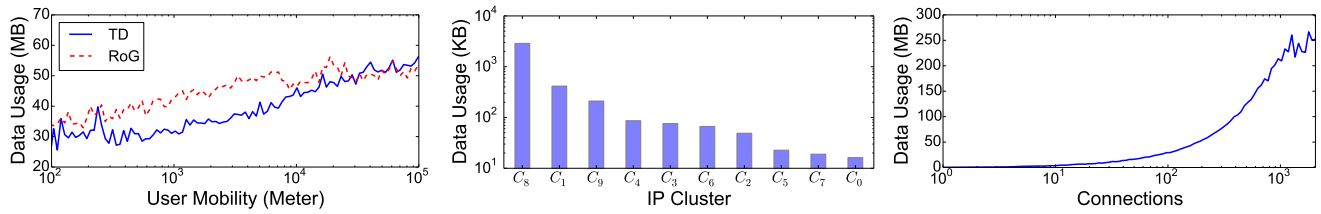
Similarly, we study the correlation between the user location and the cellular data consumption. We analyze the location with two granularities: country and point of interest (PoI). To

[2]A technology makes SIM card changeable during the usage.



(a) Cellular usage vs. weekday/weekend (b) Cellular usage vs. country (c) Cellular usage vs. PoI

Fig. 2. Correlation between different spatial and temporal factors and average daily data usage of individuals.



(a) Cellular usage vs. mobility (b) Cellular usage vs. IP cluster (c) Cellular usage vs. connection number

Fig. 3. Correlation between user behaviors and average daily data usage of individuals.

study the impact of locations where people use cellular data, we use Google Map APIs to estimate the geographical location and its corresponding PoI based on the cell information of each record.

In Fig. 2(b), we plot the 10%, 30%, 50%, 70% and 90% quantile of the daily data usage in different countries. The observation is that the daily cellular consumption is quite different when users are in different countries. For example, the 70% quantile of data usage in Japan is 59MB, while in Greece it is only around 20MB. The distribution of individual data usage is also not consistent in all countries: the median usage in Thailand is higher than that in France and Canada while the 90% quantile in Thailand is obviously below that in France and Canada. The observation suggests the consideration of such location features when performing prediction of individuals' cellular consumption.

Furthermore, we study how the distribution of PoIs affects the cellular usage. We count the PoIs of all the locations a user visited in one day and calculate the proportion of each PoI to roughly represent how much ratio of time a user spend in one type of location during that day. To demonstrate the correlation, we select 3 representative PoI labels including "School," "Park" and "Lodging," and investigate the correlation between the fractions of these PoI labels and the corresponding average cellular data usage in different locations. In Fig. 2(c) we observe that the PoI fractions also affect the average daily cellular usage. For example, as the fraction of PoI "School" increases from 0 to 0.5, the average cellular data usage drops from 60MB/day to about 20MB/day. The observation also suggests the usage of the more semantic PoI information in the prediction.

D. Mobility versus Cellular Data Usage

Next, we study the correlation between users' mobility patterns in one day and the cellular data usage. As we have all the cells' information a user passed, we are able to build his/her rough moving track. In particular, we use "trip distance (TD)" and "Radius of Gyration (RoG)" to capture the mobility of a user in one day. TD represents the accumulated moving distance a user travels in one day, and RoG has been used to capture the rough range of human mobility in several previous studies [8], [9]. We define TD and RoG as

$$\text{TD} = \sum_{i=1}^{n-1} d(l_{i+1} - l_i) \tag{1}$$

$$\text{RoG} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} d(l_i - l_{\text{center}})} \tag{2}$$

where $l_i \in \{l_1, l_2, \ldots, l_n\}$ is the location of a user has visited and $d(l_i - l_j)$ is the distance between l_i and l_j. l_{center} is the geographical center of $\{l_1, l_2, \ldots, l_n\}$, i.e., $l_{\text{center}} \equiv \frac{1}{n}\sum_{i=1}^{n} l_i$. The rationale behind the definition of RoG is that a small RoG indicates that the user has visited locations that are close to each other in a small range.

In Fig. 3(a), we present the average cellular data usage versus the user mobility. We observe that greater TD generally leads to higher cellular data usage, indicating that the longer distance a user travels in one day, the more cellular data s/he may consume, with a correlation coefficient of 0.33. Similar trend can also be observed between RoG and data usage, which shows a positive correlation between traveling range and data consumption.

E. Online Behaviors versus Cellular Data Usage

As the cellular data consumption is usually a result of users using different mobile applications, we study how users' online behaviors affect the data usage. We use both the IP addresses appearing in the connections and the number of the connection records per day to capture their online behaviors. The reason is that IP addresses can roughly reflect what applications a user uses, while the number of connections may roughly reflect how many times s/he uses them.

First, we use a data usage pair (up, down) to represent each IP address, where up (resp. down) is the average data uploaded to (resp. downloaded from) the IP address by different users in one day. We use *K-means* to cluster all the pairs of the IP addresses in the dataset into 10 clusters. We use $\{C_0, C_1, \ldots, C_9\}$ to denote these clusters, and each user's connections in one day can then be represented by a 10-dim vector according to the fractions of clusters with regard to IPs s/he connects in one day. Fig. 3(b) shows the cellular data usage of users with different IP cluster fractions. We observe that the IP distribution (mobile application patterns) also has certain impact on the cellular data usage. For example, cellular data usage of clusters C_8 is about 400 times of that for clusters C_0 and C_7.

Similarly, we investigate the correlation between daily cellular data usage and the number of connections established by a user. As illustrated in Fig. 3(c), the correlation is quite clear: more connections generally lead to more cellular data consumption, with a large correlation coefficient of 0.61, which is quite consistent with our intuition.

To summarize, this section presents our measurement studies into the individual usage of cellular data. Based on our measurement insights, we will design an individual cellular data usage prediction model for optimal cellular resource allocation in virtual network operation.

III. Individual Data Usage Prediction

Based on the insights of cellular data measurement in the previous section, we present our individual data usage prediction model.

A. Deep Prediction Model

As observed in Section II, the individual cellular data usage is affected by many confounding factors. To capture such relationship and take advantage of our large dataset, we choose to use deep learning models against generalized machine learning models and conventional linear models. Inspired by [5], we propose a wide & deep neural network (WDNN) as our data usage prediction model. As showed in Fig. 4, our proposed WDNN consists of two parts: the wide part and the deep part. The wide part is a relatively shallow network for the input of non-sequential factors such as user preference, and the deep part contains a LSTM network to capture the correlation of sequential data such as historical data usage. The rationale is to fully utilize the confounding factors measured in the previous section.

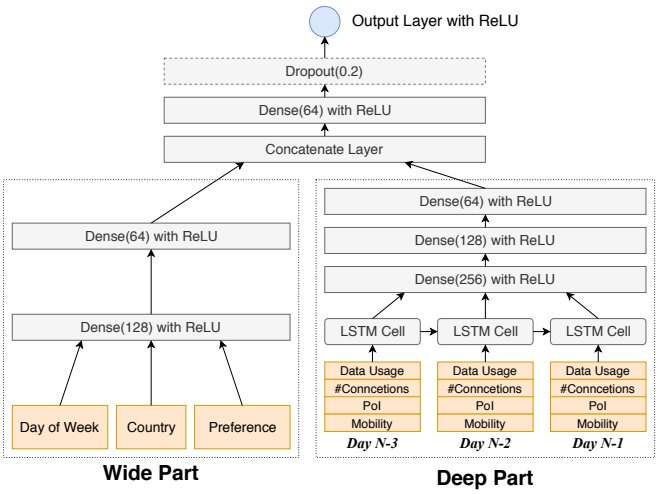


Fig. 4. Detailed structure of our WDNN.

B. Model Structure

Network structure: As illustrated in Fig. 4, the wide part of our WDNN consists of two fully-connected (dense) layers with 128 and 64 units, and the deep part has a LSTM structure with 3-day lookback and 3 following fully-connected layers. A concatenate layer is applied to join the output of two parts. Finally a dropout layer with a probability of 0.2 is added to relieve our model from overfitting. We use the $\tanh$ activation function and the hard sigmoid activation function for LSTM, and *ReLU* function for the fully-connected layer. In our training process, the mean-square error is used for loss function computation and RMSProp algorithm with a learning rate of 0.005 is used for optimization.

Input: As there are many categorical features in our dataset such as country, we choose *one hot encoding* to map the original categorical value to a high-dimension space for our WDNN inputs. For example, "Monday" is transformed into a 7-dim vector: $[1, 0, 0, 0, 0, 0, 0]$. The other continuous features are scaled into $[0, 1]$ to avoid feature imbalance for model training. The input of the wide part consists of 1) a 7-dim vector representing day of the week; 2) a 151-dim vector representing the country where the user is at; and 3) a 10-dim user preference vector based on our IP address clustering in the previous section. The input of each LSTM represents the context feature of a historical day, and it contains 1) a 2-dim vector including the historical data usage and connection count; 2) a 90-dim PoI distribution vector; and 3) a 2-dim user mobility vector including TD and RoG.

Output: Normally neural networks aim at solving classification problems. We revise the activation function of the output layer to *ReLU* function so that the output can be a non-negative continuous value, i.e., the predicted cellular data usage on the next day.

C. Prediction Performance

We verify the effectiveness of our prediction model by randomly picking 500K users from our dataset, who contributed

TABLE II
PREDICTION MODEL SCORES

Metrics (MB)	WDNN	SAPD	SVR	LSTM
MAE	43.86	52.85	67.31	50.41
RMSE	97.06	122.95	154.21	108.98

around 4.6 million samples. For model training and testing, 10-fold cross-validation is applied for precise model evaluation. We compare our WDNN model with three baseline algorithms as follows. 1) "Same as Previous Day (SAPD)," a naive baseline assuming a user will consume the same amount of data as that in the previous day; 2) "Support Vector Regression (SVR)," a generalized non-linear machine learning model with the input from the wide part only in WDNN; and 3) LSTM, a deep recurrent network with time-series input from the deep part only in WDNN. *Mean absolute error* (MAE) and *root-mean-square error* (RMSE) are used as evaluation metrics and the results are listed in Table II.

Results in Table II show that our WDNN model has the best performance, which outperforms SAPD, SVR and LSTM by 21.1%, 37.1% and 10.9% on RMSE respectively. SVR shows the poorest performance because it only considers the intrinsic factors without context features. Comparing to SAPD, a relatively better result of LSTM indicates that incorporating more context features such as user mobility can better predict user's data usage. In brief, our WDNN shows the power of combining intrinsic factors and context features to yield the best performance than naive baselines and generalized models.

IV. INDIVIDUAL DATA PLANNING: PROACTIVE MATCHING DESIGN

In this section, we present our proactive matching framework based on the formulation and analysis of individual data plan allocation problem.

A. Problem Statement

We first list the terms used in our problem statement.

1) Virtual Cellular Data Plan: In the individual data plan allocation, we assume that the virtual network operator provides a set of cellular plans, denoted as $\mathcal{P}$, for its users. Let $p_i \in \mathcal{P}$ denote each plan and r_i denote the capacity of plan p_i, i.e., the capacity in megabytes per day of the data plan.

2) User Utility and Cost Matrix: Let $\mathcal{U}$ denote the users in the system who are waiting for cellular data allocation. Each user $u_j \in \mathcal{U}$ has a particular cellular data demand denoted as d_j. To capture how good a data plan is for a certain user, we choose a utility function $f_j(p_i, d_j)$ to represent the satisfaction when a data plan p_i is allocated to user u_j according to his/her demand d_j. Examples of common utility function [10], [11] are illustrated in Fig. 5. The rationale is that large cellular data capacity will improve the user's satisfaction, but it saturates after certain intrinsic demand of the user, because people generally have limited time to spend online [11]. In real-world scenarios, d_j may contains heterogeneous factors more than the capacity of plan p_i. In fact, our design accepts any type of utility function as long as it is evaluated in $O(1)$ complexity.

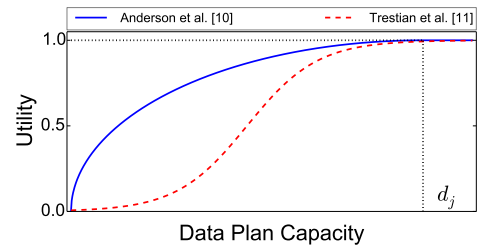


Fig. 5. Generally a user's utility increases with a larger data plan capacity, and saturates after the capacity exceeds his/her real data usage demand.

We further build a cost matrix between users and cellular data plans. In particular, we define a *cost matrix* $\mathbf{C} = \{c_{ij}\}$, where each element is defined as $c_{ij} \equiv 1 - f_j(p_i, d_j), \ \forall\, p_i \in \mathcal{P}, \ u_j \in \mathcal{U}$. A greater element c_{ij} indicates a higher cost and a lower utility between user's demand d_j and data plan p_i.

B. Problem Standardization

1) Formulation of the Problem: First, we build an *allocation matrix*, $\mathbf{A} = \{a_{ij}\}$, to represent how cellular data plans are allocated to different users. If plan p_i is allocated to user u_j, $a_{ij} = 1$, otherwise $a_{ij} = 0$ for all $p_i \in \mathcal{P}$ and $u_j \in \mathcal{U}$.

The goal of this problem is to find a specific allocation matrix minimizing the total cost defined as $\sum_{p_i \in \mathcal{P}} \sum_{u_j \in \mathcal{U}} c_{ij} a_{ij}$, i.e., maximizing the total utility of all users. Moreover, each cellular data plan can be allocated to only one user and each user can be allocated with exactly one data plan. Without loss of generality, we assume that the number of data plans is larger than the number of users, and we have the constrains as follows:

$$\sum_{u_j \in \mathcal{U}} a_{ij} = 1, \ \ \forall p_i \in \mathcal{P}. \ \ \sum_{p_i \in \mathcal{P}} a_{ij} \leq 1, \ \ \forall u_j \in \mathcal{U}. \quad (3)$$

The plan-user allocation problem can then be formulated as an optimization problem as follows:

$$\begin{aligned} \underset{\mathbf{A}}{\text{minimize}} \quad & \sum_{p_i \in \mathcal{P}} \sum_{u_j \in \mathcal{U}} c_{ij} a_{ij} \\ \text{subject to} \quad & (3), \ a_{ij} \in \{0, 1\}. \end{aligned}$$

2) Symmetric Linear Assignment: This optimization problem is an asymmetric assignment problem, which can be solved by extending it to a symmetric linear assignment problem by adding dummy variables according to Burkard *et al.* [12].

Let n and m denote the dimensions of $\mathcal{P}$ and $\mathcal{U}$, respectively. If $m < n$, we add $n - m$ users to $\mathcal{U}$ and variables $\hat{\mathbf{A}} = \{\hat{a}_{m+1}, \ldots, \hat{a}_n\}$ to $\mathbf{A}$. Furthermore, we also need to add $\hat{\mathbf{C}} = \{\hat{c}_{m+1}, \ldots, \hat{c}_n\}$ to $\mathbf{C}$ and initiate the elements $c_{ik} = 0, \ \forall i = 1, 2, \ldots, n, \ k = m+1, \ldots, n$.

Finally, the original problem can be converted to a new standard linear sum assignment problem as follows:

$$\begin{aligned} \underset{\mathbf{A}}{\text{minimize}} \quad & \sum_{i=0}^{n}\sum_{j=0}^{n} c_{ij}a_{ij} \\ \text{subject to} \quad & \sum_{j=1}^{n} a_{ij} = 1, \quad \forall i = 1,2,\ldots,n. \\ & \sum_{i=1}^{n} a_{ij} = 1, \quad \forall j = 1,2,\ldots,n. \\ & a_{ij} \in \{0,1\}. \end{aligned}$$

C. Optimal Matching Solution

1) Classical Algorithm: After transforming the allocation problem to a standard linear sum assignment problem, we can solve this with the classical Hungarian method [13], which is the first polynomial-time primal-dual algorithm based on maximum flow. Although the Hungarian method has improved the complexity of optimal solution from $O(n^4)$ to $O(n^3)$, where n is the size of the cost matrix, solving a 3000×3000 dense cost matrix on a modern CPU takes **more than** 15 minutes on average. Since the number of users and data plans can be thousands in real world, such long computation time is unacceptable in practice. Therefore a faster alternative method is needed to alleviate such pain.

Jonker and Volgenant proposed a shortest augmenting path algorithm for the linear assignment problem (LAPJV) [7] based on the Dijkstra's shortest path method. The basic idea is to consider the problem as a minimum cost flow problem and solve it by finding the shortest paths on an auxiliary graph. Although its worst case complexity is still $O(n^3)$, LAPJV can generally achieve a better performance on both dense and sparse matrices according to their experiments. We implement the Hungarian and LAPJV algorithms in Python and run several contrast experiments to re-evaluate their performance on a modern 8-core 3.7 GHz CPU setup.

2) Performance Analysis: In this experiment, we randomly generate cost matrix with different size and sparsity (each for 100 times), and measure the average computation time of the Hungarian and LAPJV algorithms. The results are listed in Table III.

The results show that 1) computation time increases with the growth of cost matrix size in an exponential pattern as the $O(n^3)$ complexity indicates; 2) Lower sparsity of the cost matrix results in longer computation time, and a fully dense problem can greatly reduce the performance of both algorithms; 3) LAPJV outperforms the Hungarian algorithm on both sparse and dense problems.

As illustrated in Fig. 2(a), 90% of total users consume less than 100 megabytes of data every day, and such light usage can be satisfied by most of the data plans offered by the virtual network operator in practice. That is, around 90% of the elements in the cost matrix is zero, i.e. $sparsity = 0.9$, and LAPJV algorithm can instantly solve a 3000×3000 sized problem within 0.1 seconds, which saves **96%** proportion of time comparing to 2.4 seconds by the Hungarian algorithm.

TABLE III
AVERAGE COMPUTATION TIME (MILLISECONDS)

Problem size	Matrix sparsity	LAP algorithm Hungarian	LAPJV
200	0.9	12.42	0.19
	0.7	13.65	0.24
	0.5	15.55	0.29
	0.0	247.98	1.68
500	0.9	91.06	1.41
	0.7	107.59	1.57
	0.5	143.55	1.89
	0.0	5.27×10^3	14.47
1000	0.9	402.15	5.58
	0.7	514.90	6.22
	0.5	1.13×10^3	8.43
	0.0	7.36×10^4	75.39
3000	0.9	2.43×10^3	96.91
	0.7	1.02×10^4	106.41
	0.5	2.03×10^4	118.38
	0.0	$> 10^6$	1.23×10^3

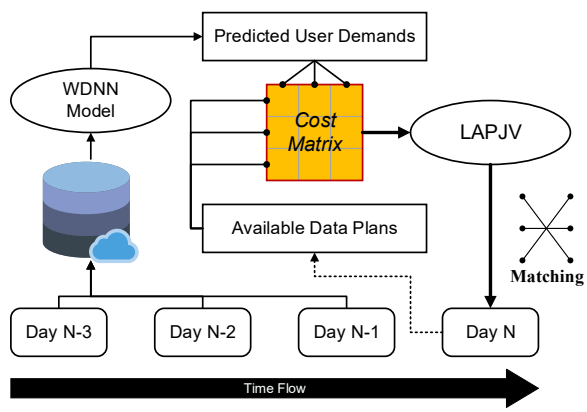


Fig. 6. Practical implementation of our CAPITAL framework.

D. Practical Implementation

Fig. 6 illustrates the practical implementation of our CAPITAL framework from the perspective of virtual network operator. For each day's allocation, we first extract the historical information, including context features and past data usage, of each user; by inputting these features into the WDNN prediction model, we have the predicted user data usage list. Then we build the cost matrix based on the pre-defined utility function, available data plans and predicted user demands. The LAPJV algorithm can instantly solve this matrix and give out an optimal matching between users and data plans. Finally the virtual network operator distributes the data plans to the assigned users online.

V. TRACE-DRIVEN EXPERIMENTS

In this section, we carry out several experiments based on our real-world data traces over 60K users in 3 selected countries. We show the effectiveness of both our prediction model and the allocation algorithm.

A. Experiment Setup

We design a simulator to run different cellular data plan allocation strategies for a virtual cellular operator, which serves users' cellular requirements that are extracted from the traces in one month. In real world, there are some limitations for network resource allocation. For example, data plans bought from ISPs in France cannot be economically allocated to users in China. Therefore we shrink our scope to country's level and select 3 typical countries from our dataset: France, Japan and Australia. We generate users and assign a cellular data usage according to the corresponding records from the one-month traces in the 3 countries.

In our experiments, we set up only one type of data plan which offers 2 gigabytes of data quota every month. The data plans are purchased by virtual cellular operator at the beginning of the month and then assigned to users according to different allocation schemes.

1) Baselines: We compare our CAPTIAL framework with the following baselines in our experiments. 1) *Monthly fixed*, in which a user is assigned with a fixed data plan during the whole month. This "inflexible" scheme is widely used by today's conventional ISPs. 2) *Daily random*, in which users are assigned with random cellular plans everyday without consideration to their demand.

2) Metrics: In our experiments, we mainly focus on average user utility as our evaluation metric. Without loss of generality, we define the user utility function as a simple $Sigmoid$-like function considering data plan capacity only: $f_{ij} = (1+\exp(6-12r_j/d_i))^{-1}$, where r_i denotes the capacity of data plan i and d_j denotes the data demand of user j, i.e., the actual data usage.

B. Experiment Results

1) Utility Gain: First, we study how our design outperforms conventional schemes, by comparing the utilities achieved at users. As illustrated in Fig. 7, we compare the average user utility achieved in the three countries during one month. We set up a moderate number of data plans bought in advance for each country, which is inferred from historical information. Our observations are as follows. First, in all cases, monthly fixed scheme has the worst performance, because newly joined user cannot receive service due to the lack of data plans. This shows the importance of a more flexible and fine-grained allocation strategy. Second, our design achieves the best utility performance and the reason is that understanding users' data usage is critical for the matching. Third, compared with the daily random scheme, our design still outperforms by 3.85% on "area under a curve", suggesting the effectiveness of our context-aware prediction and proactive allocation.

Though we set up only one type of data plan in the beginning, as time goes on the data plans are consumed to different levels and thus it is possible to match different data plans to users with different demand. The distribution of remaining data plan capacity is illustrated in Fig. 8. It shows that a) monthly fixed scheme wastes much data plan capacity though the user utility is very low, which means allocate data plans in monthly granularity is far less efficient than daily re-distribution; b) compared with our CAPTIAL scheme, the daily random scheme achieves a lower average utility because it uses the data plans less efficiently, which results in a relatively higher performance at the end of the month as it "saves" more data capacity.

2) Impact of Available Data Plans: To further study our design's performance under different amount of data plan capacity bought beforehand, we plot the monthly average utility of users achieved against the number of available data plans in Australia, which is illustrated in Fig. 9. We observe that monthly fixed, daily random and our design can all achieve an high utility level of 0.95, but the corresponding data plan is 7000, 2200 and 1500 respectively. The results suggest that our design can achieve a good utility with much lower costs. In particular, our design can save the virtual network operator's cost by up to 31.8% and 78.6% against daily random and monthly fixed schemes respectively, for the same average utility level of 0.95.

3) Impact of Prediction: Experiments above have verfied the effectiveness of our allocation algorithm. To fairly evaluate the impact of prediction models on our framework, we "predictively" set the number and capacity of daily data plans exactly the same as the real data usage of users on that day. Therefore a "perfect" prediction model along with the optimal allocation algorithm should give a precise matching and the average utility is just 1.0. We evaluate different prediction models on the traces in Australia in one month and the results are showed in Fig. 10. It shows that our WDNN prediction model can achieve the best average utility, which outperforms SVR by 19.8%.

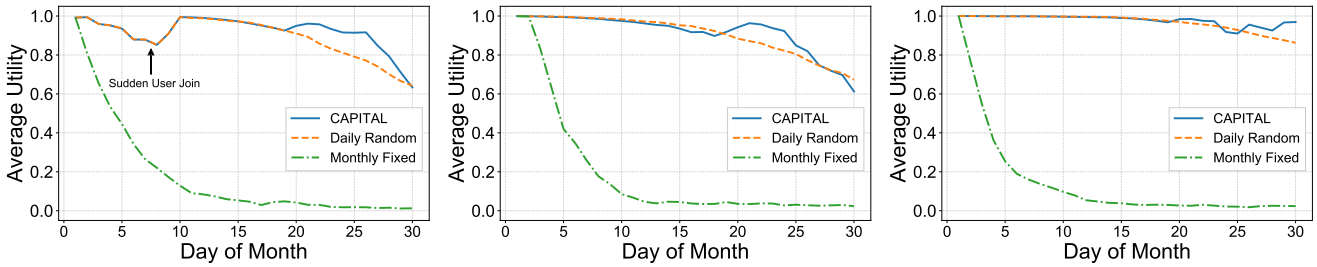
4) Individual Insights: From the perspective of individual users, we consider a user being "unsatisfied" when the allocated data plan cannot meet with his/her cellular data demand, i.e., utility is less than 1.0. We plot the unsatisfied proportion of users with different data demand in Fig. 11 from one-month traces in France. We observe that a larger data demand leads to a higher probability of being unsatisfied because those users' demand is underestimated by our perdiction model. Though our prediction model fairly treats "underestimation" and "overestimation" during training, underestimation can be more critical than overestimation in practice. Meanwhile, users with relatively small data demand (i.e., less than 10MB), are also more likely to be neglected, which is a weakness of our framework.

VI. Related Works

In this section, we present related studies on virtual network operator, individual data usage observations, cellular traffic prediction approaches and cellular network resource strategies.

A. Mobile Virtual Network Operator

Nowadays mobile virtual network operators have become popular alternatives for conventional ISPs. Zarinni *et al.* [14] first measured the performance of two common MVNO families and observed significant performance degradation. Zheng



(a) Australia ($\#Plans = 1500$) (b) Japan ($\#Plans = 3500$) (c) France ($\#Plans = 4500$)

Fig. 7. Daily average user utility trend with different allocation strategies in 3 selected countries in one month.

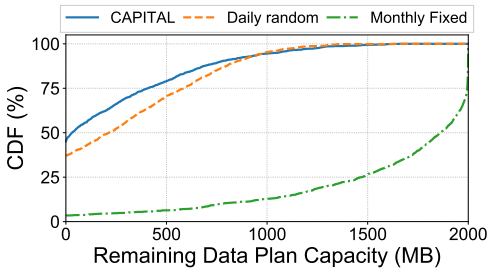


Fig. 8. Distribution of remaining data plan capacity on the last day in Australia.

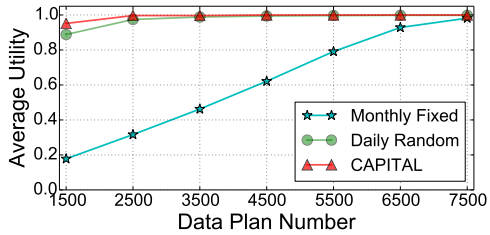


Fig. 9. Average utility of different initial data plan numbers bought by the network operator in Australia.

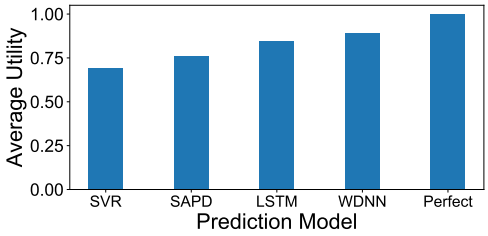


Fig. 10. Average utility of different prediction models.

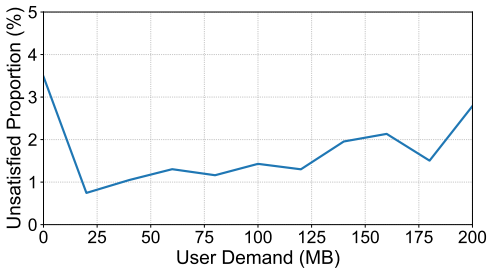


Fig. 11. Unsatisfied proportion of users with different data demand.

et al. [15] provided a systematic analysis of virtual operator's economic impact and Li *et al.* [16] developed a policy to maximize profit under dynamic network environment. These studies were mainly measurement and theoretical works with simulation experiments only.

B. Individual Mobile Network Usage Patterns

To understand the characteristics of individual mobile network usage, Paul *et al.* [17] and Oliveira *et al.* [3], [18] carried out measurement studies and revealed certain patterns: a large proportion of data traffic is generated by only a small fraction of users; daily repetitive patterns exist in individual traffic, and changes at different hours of the day. Jo *et al.* [19] also identified the spatiotemporal correlations of cellular usage patterns. Different usage patterns of smartphone users were observed under cellular and Wi-Fi networks by Fukuda *et al.* [20].

C. Cellular Traffic Prediction

Massive works have been studied on the prediction of cellular data traffic. Xu *et al.* [21] leveraged a linear prediction model to forecast the regularity component decomposed from the original cellular traffic. Li *et al.* [22] leveraged the α-stable modeled property to process the temporal variation of application-level traffic data. Wang *et al.* [23] proposed a hybrid deep learning model for spatiotemporal prediction. Zhang *et al.* [24] proposed a double spatiotemporal network architecture to forecast the traffic volume in two cities. However, existing works mainly focused on the prediction of aggregated traffic, e.g., traffic inside a cell. To the best of our knowledge, we are the first to incorporate both spatiotemporal and user behavior information for individual cellular data usage prediction.

D. Cellular Network Resource Strategies

Many studies have been carried on network resource strategies for different requirements under different environments. A

usage-based pricing model was proposed [25] to compute the optimal prices offered to the customers, which requires ISPs to solve a large scale non-linear optimization problem. A time-dependent pricing trial was presented by [26] and this strategy could reduce the need for network over-provisioning and increase profit. Recently, a dynamic pricing model with service level agreements has been proposed [27], which determines appropriate prices based on current state, ensuring clients' connection and the requested bandwidth. Rollover data plans were also proposed to allow users to transfer unused data quotas this month to the next month [28].

The limitation of previous cellular resource allocation is that they either based on straightforward rules or the assumption that users are rationale enough to pick the most suitable data plans for themselves. However, in practice, users are usually not able to determine how much capacity they may actually need in the future. Prediction of individual cellular usage is thus critical.

VII. Concluding Remarks

Always being connected to the internet has become a must for today's people and using cellular networks is inevitable as people's mobility increases. To handle the high cost of cellular data plans, mobile virtual network operator has become a promising alternative for conventional ISP. However, common strategies including naive allocation and static pricing schemes are not efficient and flexible enough for today's cellular data plan allocation. Based on large-scale measurement studies over 2 years' real-world operation traces, we reveal that individual and fine-grained cellular data plan allocation is in demand, and both spatiotemporal and user behavior patterns affect cellular usage. To this end, we propose CAPITAL, a proactive matching framework of matching cellular data plans to users with different demand. A context-aware wide & deep model is designed to capture individual data usage, and a prediction based data plan matching algorithm is applied to proactively allocate plans to users in an instant manner. Our trace-driven experiments show that our design can reduce the data usage cost by 78.6%, while retaining the same user experience level.

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